

Distance on Numeric Data: Minkowski Distance

- **Minkowski distance**: A popular distance measure

$$d(i, j) = \sqrt[p]{|x_{i1} - x_{j1}|^p + |x_{i2} - x_{j2}|^p + \dots + |x_{il} - x_{jl}|^p}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{il})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jl})$ are two l -dimensional data objects, and p is the order (the distance so defined is also called L- p norm)

- Properties

- $d(i, j) > 0$ if $i \neq j$, and $d(i, i) = 0$ (Positivity)
- $d(i, j) = d(j, i)$ (Symmetry)
- $d(i, j) \leq d(i, k) + d(k, j)$ (Triangle Inequality)

- A distance that satisfies these properties is a **metric**
- Note: There are nonmetric dissimilarities, e.g., set differences